VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI- 590 018

"AICTE Activity Point Programme"

Submitted in Partial Fulfillment for the Award of Degree of

Bachelor of Engineering
in
Electronics and Communication Engineering

Submitted by
DHANUSH S N
1CR21EC068

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Assistant Professor, Dept. of ME

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CMR INSTITUTE OF TECHNOLOGY, BENGALURU.
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AICTE Activity Point Programme CERTIFICATE

This is to certify that the **AICTE Activity Point Programme** has been successfully carried out by **DHANUSH S N** bearing **USN 1CR21EC068,** bonafide student of CMR Institute of Technology in partial fulfillment of the requirements for the award of degree of **Electronics and Communication Engineering** of **Visvesvaraya Technological University, Belagavi** during **2024-25**. It is certified that all corrections and suggestions indicated for Internal Assessment have been incorporated in the report. The AICTE Activity Point Programme report has been approved as it satisfies the academic requirements in respect of AICTE Activity Point Programme for the said degree.

……	……	……	……
Ms. Anju Das	**Dr. Pappa M**	**Dr. Narendra N**	**Dr. Sanjay Jain**
Assistant Professor	**Professor & Head**	**Assistant Professor**	**Principal**
Dept. of ECE,	**Dept. of ECE,**	**Dept. of ME**	**CMRIT**
Faculty Incharge,	**CMRIT**	**AICTE activity points Coordinator,**	
CMRIT		**CMRIT**	

DECLARATION

I **DHANUSH S N** bearing the **USN**:1CR21EC258, student of Bachelor of Engineering, **Electronics and Communication of CMR Institute of Technology, Bengaluru,** hereby declare that the **AICTE Activity Point Programme** work has been carried out by me under the supervision and guidance of Department Staff Coordinators submitted by me as a partial fulfillment for the award of **Bachelor of Engineering** degree in **Electronics and Communication Engineering** from **Visvesvaraya Technological University, Belagavi** during **2024-25**.

I hereby declared that the AICTE activity point work has been carried out at following partner organizations:

1. BHAVANI FARMS
2. NAVODAYA COMPOSITE SCHOOL, KADADANAMARI &
3. YELSANDRA GRAM PANCHAYAT,

Signature

DHANUSH S N

1CR21EC068

ACKNOWLEDGEMENT

I extend my heartfelt gratitude to the **Management Members**, along with **Mr. Sanjay Jain**, Principal of **CMR Institute of Technology**; **Asst Professor, Mrs Anju Das** Professor at **CMR Institute of Technology**; and **Dr. Narendra N**, AICTE Activity Points Coordinator at **CMR Institute of Technology**, for their unwavering support and provision of essential resources for the successful execution of this program.

I am also deeply thankful **to BHAVANI FARMS, NAVODAYA COMPOSITE SCHOOL, KADADANAMARI and YELSANDRA GRAM PANCHAYAT** for their invaluable support in offering meaningful volunteering opportunities and resources.

Lastly, I would like to express my sincere appreciation and respect to my parents, the teaching and non-teaching staff of the Department, and all my friends who have supported me, directly or indirectly, throughout my AICTE Activity Point Program journey.

DHANUSH S N
1CR21EC068

Work Content

Sl. No.	Activity Heads	From date- To date	Number of Week	Total No of Hours	Performance Appraisal / Points earned (To be Filled by Coordinators)	Remark
1	Developing And Managing Efficient Garbage Disposal System	**02-11-2024 to 12-11-2024**	**1.5**	**80**		
2	Facilitating 100% Digitized Money Transactions	**15-11-2024 to 23-11- 2024**	**01**	**80**		
3	Development And Implementation Of Sustainable Water Management System	**1-12-2024 to 10-2024**	**1.5**	**80**		
4	Contribution To Any National Level Initiative Of Government Of India	**15-10-2024 to 26-10- 2024**	**1.5**	**80**		
5	Helping Local Schools To Achieve Good Results And Enhance Their Enrolment In Higher/ Technical/ Vocational Education	**11-10-2024 to 23-10-2024**	**01**	**80**		
TOTAL ACTIVITY POINTS EARNED						

ABSTRACT

As part of community development and sustainability initiatives, several impactful activities were undertaken focusing on environmental management, digital empowerment, and educational support.

One major activity involved the development and implementation of a sustainable water management system. This included planning and executing techniques to conserve water resources, optimize usage in agriculture, and promote environmentally responsible practices.

Another key project focused on developing and managing an efficient garbage disposal system. The activity emphasized proper waste segregation, creating awareness about hygiene, and ensuring the safe and systematic disposal of waste materials to support a cleaner and healthier environment.

In the area of digital transformation, active efforts were made in facilitating 100% digitized money transactions. This involved integrating UPI systems and encouraging the use of digital platforms for secure and cashless payments, contributing to the broader vision of a digitally empowered society.

Educational outreach was also a vital part of the initiatives. Under one activity, support was provided to local schools to help students achieve better academic results and improve their enrolment in higher, technical, or vocational education. Awareness sessions, guidance programs, and motivational interactions were conducted to inspire and inform school students.

Another initiative involved contributing to national-level government program**s**, where awareness about various schemes and initiatives was spread to the local community. The goal was to ensure that more people understood and benefited from government-led developmental efforts.

These activities collectively contributed to creating sustainable solutions, enhancing digital literacy, and empowering local communities through knowledge and action.

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CERTIFICATES PHOTOCOPY

Yelesandra Gram Panchayat

Yelesandra, Kolar (D), karnataka, 563114

DATE: 12-11-2024

CERTIFICATE

This is to certify that the undersigned students from the 7th semester of the ELECTRONICS **AND COMMUNICATION ENGINEERING** department studying at **CMR INSTITUTE OF TECHNOLOGY (CMRIT), BENGALURU**, have successfully completed and activity titled **"DEVELOPING AND MANAGING EFFICIENT GARBAGE DISPOSAL SYSTEM"** at, Yelesandra, Kolar(D), Karnataka-563144. The activity was conducted **from 02-11-2024 to 12-11-2024**, as part of the AICTE initiative. We hope that this activity contributes positively to national development.

DHANUSH S N
1CR21EC068

Seal and Signature

Navodaya composite high school

kadadanamari, Chintamani(t),chikkaballapura (D), karnataka, 563123

DATE: 23-10-2024

CERTIFICATE

This is to certify that the undersigned students from the 7th semester of the ELECTRONICS **AND COMMUNICATION ENGINEERING** department studying at **CMR INSTITUTE OF TECHNOLOGY (CMRIT), BENGALURU**, have successfully completed and activity titled **"HELPING LOCAL SCHOOLS TO ACHIEVE GOOD RESULTS AND ENHANCE THEIR ENROLMENT IN HIGHER/ TECHNICAL/ VOCATIONAL EDUCATION"** at, Kadadanamari, Chikkaballapura(D), Karnataka-563123. The activity was conducted **from 11-10-2024 to 23-10-2024**, as part of the AICTE initiative. We hope that this activity contributes positively to national development.

DHANUSH S N
1CR21EC068

HEAD MASTER
Navodaya
Kadadanamari
Chintamani Tq
Karnataka

Seal and Signature

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kadadanamari, Chintamani(t), chikkaballapura (D), karnataka, 563123

DATE: 26-10-2024

CERTIFICATE

This is to certify that the undersigned students from the 7th semester of the ELECTRONICS **AND COMMUNICATION ENGINEERING** department studying at **CMR INSTITUTE OF TECHNOLOGY (CMRIT), BENGALURU**, have successfully completed and activity titled **"CONTIBUTION TO ANY NATIONAL LEVEL INITIATIVE OF GOVERNMENT OF INDIA"** at, Kadadanamari, Chikkaballapura(D), Karnataka-563123. The activity was conducted **from 15-10-2024 to 26-10-2024**, as part of the AICTE initiative. We hope that this activity contributes positively to national development.

DHANUSH S N
1CR21EC068

HEAD MASTER
Navodaya
Kadadana
Chintamani Tq
Karnataka

Seal and Signature

||Sri Muneshwara Swamy Prasana||

Prop: Ravindra Reddy

Bhavani Farms

Kuthaganahalli Village,Sarjapura Hobli,Anekal Taluk,Bengaluru-562125
8550848363 / 9036961673 / 8749008828

Date:25-Nov-2024

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **DHANUSH S N**, bearing the USN: **1CR21EC068**, has actively participated in the **Facilitating 100% Digitized money transactions** in our farms. The said activity took place **15th November 2024 to 23rd November, 2024.**

During this period, **DHANUSH S N,** demonstrated a commendable level of dedication and responsibility. His performance in the various tasks he undertook was satisfactory, showcasing his ability to achieve all our transactions digitally through our website and integrate the UPI system for our farms.

We appreciate **DHANUSH S N,** efforts and commitment towards promoting this in our farms. We believe that their involvement in these activities has not only benefited us but also contributed to broader understanding of digitalized payments.

If you require any further information, please do not hesitate to contact us.

Sincerely,
Ravindra Reddy
Bhavani Farms

For BHAVANI FARM
Proprietor

Seal & Signature

ǁSri Muneshwara Swamy Prasanaǁ

Prop: Ravindra Reddy

Bhavani Farms

Kuthaganahalli Village,Sarjapura Hobli,Anekal Taluk,Bengaluru-562125
8550848363 / 9036961673 / 8749008828

Date:12-Dec-2024

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **DHANUSH S N,** bearing the USN: **1CR21EC068** has actively participated in the **Development and implementation of sustainable water management system** activity in our farms. The said activity took place **1st December, 2024 to 10th December, 2024.**

During this period, **DHANUSH S N**, demonstrated a commendable level of dedication and responsibility. Their performance in the various tasks assigned was satisfactory, showcasing their ability to contribute effectively to sustainable practices and environmental conservation efforts.

We appreciate **DHANUSH S N,** efforts and commitment towards promoting sustainable water management. We believe that their involvement in these activities has not only benefited us but also contributed to broader environmental sustainability goals.

If you require any further information, please do not hesitate to contact us.

Sincerely,
Ravindra Reddy
Bhavani Farms

For BHAVANI FARM
Proprietor

Seal & Signature

CHAPTER 1

INTRODUCTION

This report presents a comprehensive overview of the community service and outreach activities undertaken as part of the Activity Points requirement in Yelesandra gram panchayat .These activities were designed to bridge the gap between academic learning and societal needs, encouraging students to engage in real-world problems and contribute to the betterment of society. The program serves as a platform to develop essential life skills such as leadership, teamwork, communication, social awareness, and problem-solving.

The focus areas of the activities included environmental sustainability, digital empowerment, waste management, and educational outreach. Through these initiatives, efforts were made to address pressing issues such as water conservation, effective waste disposal, promotion of digital payments in rural areas, and improving awareness of higher education among school students. Each activity was aligned with one or more national missions like Swachh Bharat Abhiyan, Digital India, and Unnat Bharat Abhiyan, contributing directly to the development and upliftment of under-served communities.

The environment activities involved hands-on work in water management and sanitation systems, aiming to build awareness and implemented eco-friendly practices in rural settings. Digital literacy efforts focused on empowering local individuals and workers with knowledge of UPI and digital transaction systems, promoting transparency and ease of use in daily transactions. The educational outreach programs aimed to inspire students from government schools to purse further education by conducting motivational sessions

Participating in these activities not only created a positive social impact but also enhanced personal growth and civic consciousness. The exposure to ground-level issues and the opportunity to contribute directly to society helped develop a sense of empathy and responsibility, essential traits for any future professional

CHAPTER 2

ACTIVITY 1

Developing and Managing an Efficient Garbage Disposal System

2.1.1 Overview

This activity was aimed at addressing the challenges of improper waste management in rural and semi-urban areas by designing and promoting an efficient and sustainable garbage disposal system. The primary objective was to create awareness about the importance of cleanliness, proper waste segregation, and hygienic disposal practices among the local community.

The initiative involved conducting a survey to understand existing waste disposal methods, identifying problem areas, and proposing improved techniques for collection, segregation, and disposal of household and agricultural waste. Emphasis was placed on separating biodegradable and non-biodegradable waste, encouraging composting for organic matter, and suggesting recycling methods for plastics and other non-degradable materials.

Interactive sessions and demonstrations were conducted to educate villagers about the health and environmental hazards of open dumping and burning of waste. Community members were also introduced to low-cost, practical solutions like compost pits, dry waste collection centers, and daily waste collection schedules.

In addition to community education, the activity involved collaborating with local authorities and volunteers to set up basic infrastructure for waste collection and disposal. Visual aids such as posters and banners were used to reinforce the message and promote long-term behavior change.

The outcome of the activity was increased awareness, improved cleanliness in public areas, and the adoption of more structured waste management practices by households. The initiative contributed to the goals of the **S**wachh Bharat Abhiyan, promoting hygiene and sustainable living in the community.

2.1.2 Efficient Garbage Disposal System Can Be Developed and Managed By

Developing and managing an efficient garbage disposal system involves a combination of planning, community involvement, awareness, and basic infrastructure. Below are the key steps involved in implementing such a system:

1. **WasteSegregationatSource**

The first and most important step is educating households and businesses to segregate waste at the source into:

- Biodegradable waste (e.g., food scraps, garden waste)
- Non-biodegradable waste (e.g., plastics, metal, glass)
- Hazardous waste (e.g., batteries, sanitary waste)

Color-coded bins can be provided or promoted (green for wet waste, blue for dry waste, red for hazardous).

2. **CommunityAwarenessandTraining**

Conducting regular awareness campaigns helps in changing the community's behavior towards waste. People need to understand why segregation matters and how to do it correctly. Training sessions and demos can help illustrate composting, recycling, and proper disposal methods.

3. **CompostingOrganicWaste**

For biodegradable waste, community or household-level compost pits or compost bins can be introduced. Organic waste naturally breaks down into compost, which can be used as fertilizer in gardens or farms. This reduces the amount of waste needing external disposal.

4. **DryWasteCollectionandRecycling**

Non-biodegradable materials like plastics, paper, and glass can be collected weekly and sent to recycling centers. Partnerships with scrap dealers or local recycling units can ensure proper processing and reuse of this waste.

5. **CentralizedWasteCollectionPoints**

Small villages or localities can set up designated collection points or bins for different waste types. These should be easily accessible and cleaned regularly. Local volunteers or workers can be assigned to collect waste daily or on alternate days.

6. **CollaborationwithLocalAuthorities**

Local government bodies or panchayats can help with resources like bins, transportation of waste, and linking to municipal waste management facilities. A small team can be appointed to monitor and manage the process.

7. **WasteCollectionSchedule**

A fixed schedule (e.g., wet waste collected daily, dry waste twice a week) should be followed and displayed in common areas. This brings discipline and routine to waste management.

8. **MonitoringandMaintenance**

Continuous monitoring is required to ensure the system is followed. Community volunteers or student groups can help track participation, identify issues, and suggest improvements.

9. **RewardandRecognition**

Motivating the community by recognizing households or individuals who follow best practices encourages others to join. Simple rewards or public appreciation can boost participation.

By following these steps, even small communities can maintain a clean, healthy environment and contribute to sustainable waste management practices. It also supports national goals like the Swachh Bharat Abhiyan, promoting cleanliness and environmental responsibility at the grassroots level.

ACTIVITY 2

Facilitating 100% Digitized Money Transactions

2.2.1 Overview

Promoting digital transactions is a vital step toward achieving financial inclusion and building a cashless economy so we implemented and helped Bhavani farms to 100% digitized transactions. This activity aimed to facilitate the complete adoption of digitized money transactions in rural and semi-urban areas where cash-based systems are still widely used. The initiative focused on educating individuals and small business owners about the benefits and usability of digital payment platforms, thereby empowering them to carry out secure and efficient financial transactions.

The primary target group included farmers, small-scale vendors, shopkeepers, and self-employed workers who were either unaware of or hesitant to use mobile-based payment systems. Through interactive awareness campaigns and one-on-one guidance, the activity introduced participants to user-friendly digital tools such as UPI apps (e.g., PhonePe, Google Pay, BHIM, Paytm), mobile wallets, and QR code-based payment methods.

Hands-on training sessions were conducted to demonstrate how to download and install UPI apps, link bank accounts, set up secure UPI PINs, and initiate transactions. Participants were also trained to generate and use QR codes for receiving payments, especially useful for vendors and shopkeepers. Special care was taken to educate users on digital safety practices such as recognizing phishing attempts, avoiding sharing OTPs, and regularly monitoring account activity.

One of the key outcomes of this activity was the increased confidence among users in handling digital financial tools. Vendors who previously relied only on cash started accepting UPI payments, resulting in faster, safer, and more traceable transactions. The shift also reduced dependency on physical currency, encouraged transparency, and aligned with national initiatives such as Digital India and Financial Literacy Programs.

This initiative not only helped individuals gain digital competency but also contributed to broader economic transformation by connecting underserved communities to the mainstream financial

ecosystem. It demonstrated how technology can be effectively leveraged to simplify daily financial tasks and improve overall quality of life.

2.2.2 Implementation Strategy

Implementing a 100% digitized money transaction system in rural or semi-urban areas involves a step-by-step approach that blends awareness, accessibility, and practical training. The first step is to identify the target community where digital financial practices are limited. This could include small vendors, daily wage workers, farmers, and local shopkeepers. An initial survey or field visit helps in understanding the current level of digital literacy and the barriers preventing adoption—such as fear of technology, language barriers, or lack of trust in online transactions.

The second step involves raising awareness through workshops and demonstrations. Interactive sessions can be organized in local community centers, markets, or schools to explain the benefits of digital transactions—such as safety, speed, transparency, and convenience.

Next, practical implementation support should be provided. This includes helping local vendors set up QR codes for their businesses, offering troubleshooting support for app-related issues, and encouraging customers to use UPI instead of cash. Digital banners, stickers, and posters with QR codes and instructions can be displayed prominently at shops and stalls to increase visibility and usage. In addition, handouts or guides in the local language can be distributed for future reference.

A crucial part of the process is educating the community on digital safety and fraud prevention. Users must be taught to avoid sharing OTPs, passwords, or personal information, and how to report suspicious activity.Finally, the system must be monitored and supported with follow-up visits. Regular feedback should be collected to understand any challenges faced by users. Local volunteers or student coordinators can help track usage, resolve issues, and ensure long-term adoption. Over time, as trust builds and familiarity grows, more users will transition to digital modes of payment, creating a sustainable and inclusive financial environment

ACTIVITY 3

Development and Implementation of Sustainable Water Management System

2.3.1 Overview

The activity undertaken at Bhavani Farms focused on addressing the growing need for sustainable water use in agriculture by developing and implementing an efficient and eco-friendly water management system. With increasing pressure on natural water resources and the erratic nature of rainfall due to climate change, the goal of this initiative was to reduce water wastage, improve irrigation efficiency, and ensure the long-term availability of water for farming operations.

The project began with a thorough analysis of the farm's existing water sources, such as borewells and open wells, and current irrigation practices, which primarily involved traditional flood irrigation. It was observed that a significant amount of water was being lost due to evaporation, seepage, and over-irrigation, leading to reduced soil fertility and waterlogging in some areas. To overcome these challenges, a comprehensive sustainable water management plan was designed specifically for the conditions at Bhavani Farms.

One of the major steps was the implementation of a drip irrigation system, which delivers water slowly and directly to the base of each plant. This not only minimizes water loss but also promotes better plant health and crop yield. The team also assisted in constructing rainwater harvesting structures, including catchment areas and percolation pits, to collect and store rainwater effectively during the monsoon season. This water could then be reused during dry months, reducing dependence on groundwater and promoting natural aquifer recharge.

In addition to infrastructure development, the project emphasized education and capacity-building. Farm workers and caretakers were trained in system maintenance, scheduling of irrigation based on crop needs, and monitoring soil moisture levels using basic tools. Discussions were also held on water budgeting and how to track usage for different agricultural activities to further enhance efficiency.

The implementation of this sustainable system brought multiple benefits. It led to a significant reduction in water consumption, stabilized groundwater usage, and improved overall crop

productivity. It also helped reduce operational costs by cutting down on electricity usage for pumping and minimizing water wastage. The success of this model has set an example for nearby farms and small landholders, showing that even with limited resources, sustainable and smart water management is possible.

Overall, the initiative at Bhavani Farms is a practical demonstration of how agricultural practices can be aligned with environmental sustainability. It highlights the importance of combining modern techniques with local knowledge to protect vital resources like water while enhancing productivity and resilience in farming.

2.3.2 Development of the Sustainable Water Management System.

The sustainable water management system developed at Bhavani Farms functions through a combination of drip irrigation, rainwater harvesting, and efficient water distribution methods. At its core, the system replaces conventional flood irrigation with a drip-based approach, where water is delivered directly to the base of each plant through a network of pipes and emitters. This method ensures minimal water loss due to evaporation or runoff and provides moisture exactly where it is needed, leading to healthier plant growth and improved crop yields.

The water used in the drip system is sourced from both existing borewells and rainwater harvesting pits constructed across the farm. These pits collect rainwater during the monsoon season, which is stored either in tanks or allowed to percolate into the ground, contributing to groundwater recharge. A simple manual control mechanism using valves and filters allows farm workers to regulate the amount and timing of water flow to different parts of the field based on crop requirements.

The system is low-maintenance and user-friendly, enabling farmers to adjust water delivery with ease while avoiding over-irrigation. Periodic checks are conducted to clean filters, maintain emitters, and ensure uninterrupted flow throughout the irrigation network. This integrated approach promotes efficient resource use and significantly reduces the energy needed for pumping and distribution.

2.3.3 Benefits of the System

Water Conservation: The most significant benefit is the substantial reduction in water usage. Drip irrigation uses up to 60–70% less water compared to traditional methods, allowing the farm to operate efficiently even during periods of low rainfall.

Improved Crop Health: By providing consistent and precise watering, plants receive optimal moisture levels, leading to better root development, higher nutrient absorption, and increased yields.

Cost Efficiency: Reduced water consumption and lower power usage for pumping have directly led to a decrease in operational costs. Over time, the investment in drip and rainwater harvesting infrastructure is offset by savings on electricity and water.

Soil Protection: Unlike flood irrigation, drip systems prevent waterlogging, reduce erosion, and maintain soil structure, which contributes to long-term soil fertility and productivity.

Sustainability and Groundwater Recharge: The rainwater harvesting structures help in capturing and storing rainwater, thereby reducing the burden on groundwater sources and contributing to natural aquifer recharge.

Ease of Use and Replicability: The system is simple to operate and requires minimal technical expertise, making it easily replicable across similar farm setups. Farm workers were trained to manage the system independently, ensuring long-term functionality.

Environmental Impact: By reducing water and energy consumption, the system contributes to the broader goals of sustainable farming and environmental conservation. It aligns well with modern agricultural practices aimed at resource optimization.

ACTIVITY 4

Contribution To Any National Level Initiative Of Government Of India

2.4.1 Overview

This activity was conducted as part of an effort to contribute to a national-level initiative launched by the Government of India, with a focus on spreading awareness and promoting civic responsibility among school students. The selected venue for this initiative was Navodaya School in Kadadanamari**,** a rural institution where students were introduced to key national campaigns through interactive sessions and hands-on participation.

The primary objective of the program was to educate students about flagship government schemes such as Swachh Bharat Abhiyan (Clean India Mission)**,** Digital India, and Fit India Movement**,** and to encourage them to become active participants in these campaigns. The activity was planned to align with the broader national vision of creating informed, responsible, and proactive youth citizens.

During the visit, interactive presentations and group discussions were held with students to explain the purpose and importance of these initiatives in simple, relatable terms. As part of Swachh Bharat Abhiyan**,** **s**tudents participated in a school campus cleanliness drive, learning about hygiene, waste segregation, and the long-term impact of sanitation on public health. Posters and placards were used to convey key messages visually, making the session engaging and easy to understand.

For Digital India**,** the focus was on introducing the concept of digital services, the benefits of online platforms, and basic cybersecurity tips. Students were encouraged to explore safe and constructive uses of the internet, especially in learning and communication.

Under the Fit India Movement**,** students were guided through simple physical exercises, yoga routines, and discussions about daily habits that promote mental and physical well-being. A mini fitness challenge was conducted to encourage participation and promote a healthy lifestyle.

The activity not only created awareness but also empowered students to carry these messages home, acting as ambassadors of change within their families and communities. It showcased how educational

outreach programs can effectively contribute to national missions by sowing the seeds of social and civic responsibility at a young age.

2.4.2 Personal Contribution to the Initiative

As part of my contribution to a national-level initiative, I actively participated in organizing and executing an awareness and outreach program at Navodaya School, Kadadanamari. The activity was aimed at introducing school students to key Government of India missions, including Swachh Bharat Abhiyan, Digital India, and the Fit India Movement. My role involved both behind-the-scenes planning and direct interaction with students to ensure the success of the program.

I began by helping to design the structure of the sessions, including preparing a simple, informative presentation and selecting appropriate themes for each initiative. I created visually engaging posters and flashcards with slogans, facts, and illustrations that were displayed during the sessions to grab attention and enhance learning. These materials were especially helpful in bridging the knowledge gap and maintaining the interest of younger students.

During the Swachh Bharat Abhiyan segment, I led a cleanliness drive around the school premises. I demonstrated how to properly collect and dispose of litter, educated the students on the importance of sanitation, and helped them understand the health hazards associated with unhygienic surroundings. I also assisted in forming small student groups to take up responsibilities for cleaning specific areas, encouraging teamwork and a sense of ownership.

In promoting the Digital India campaign, I delivered an interactive session on digital literacy. I explained how digital platforms are transforming services such as education, banking, and communication. I introduced students to basic smartphone features, discussed online safety measures, and emphasized the benefits of using digital platforms responsibly. For senior students, I briefly explained UPI, online payments, and the role of government apps and portals like DigiLocker and UMANG.

As part of the Fit India Movement, I conducted a brief morning fitness session where students were guided through basic warm-up exercises, yoga asanas, and breathing techniques. I also organized a

quick quiz on health and fitness awareness to make the session more engaging. The aim was to make physical activity fun while promoting a culture of regular exercise among the students.

In addition to conducting the sessions, I engaged in one-on-one interactions with students, answering their questions and helping them relate the national campaigns to their daily lives. I also coordinated with teachers to encourage them to carry forward the momentum by integrating these themes into regular classroom discussions and school activities.

Overall, my contribution was focused on making the initiative informative, participatory, and impactful. By actively engaging students and providing them with knowledge, tools, and motivation, I helped spark a sense of civic awareness and encouraged them to be contributors to these important national missions. This experience also gave me the opportunity to develop leadership, communication, and organizational skills while giving back to the community in a meaningful way.

ACTIVITY 5

Helping Local Schools To Achieve Good Results And Enhance Their Enrolment In Higher/ Technical/ Vocational Education

2.5.1 Overview

This activity was carried out with the intention of making a meaningful impact on the academic development of students at Navodaya School, Kadadanamari, particularly focusing on those in secondary and higher secondary levels. The core objectives of the program were twofold: to help students achieve better academic results through strategic mentoring and academic support, and to create awareness about opportunities in higher education, technical training, and vocational courses.

The activity began with an interactive survey and conversations with students and teachers to understand the key academic challenges faced by the school. Based on the inputs received, a plan was developed to provide both academic guidance and career awareness support. The sessions targeted students from Class 8 to Class 10, many of whom were unaware of the diverse opportunities available after their school education.

Academic mentoring sessions focused on concept clarity in core subjects such as Mathematics, Science, and English. Students were guided through important concepts, previous year question patterns, and model papers. These sessions were designed to build foundational understanding, improve problem-solving skills, and provide personalized attention to those struggling with specific topics. Study tips, memory techniques, and time management strategies were also shared to help them prepare more effectively for board exams.

Beyond academics, a major part of the initiative was to broaden students' perspectives about their future education options. Information sessions were conducted on various streams available after Class 10 and 12, including Intermediate education, Polytechnic diplomas, ITI (Industrial Training Institutes), National Skill Development Programs, Engineering and Medical entrance exams, and government-run programs like PMKVY (Pradhan Mantri Kaushal Vikas Yojana).

These sessions were presented in a simple and relatable way, using charts, visuals, and real-life examples. Students were also informed about scholarship schemes such as NSP (National Scholarship Portal), Vidya Lakshmi, and state-level merit-based scholarships that could help ease the financial burden of higher education.

In addition, career guidance talks were held to explain the importance of setting goals and planning ahead. Students were encouraged to explore their interests and identify potential career paths based on their strengths. A few students were even given personalized mentoring on how to apply for specific entrance exams, including practical steps like registration, exam structure, and preparation tips.Teachers at the school were also engaged in discussions to ensure continuity of these efforts, and learning materials were shared with them for future use. This helped in building a sustainable model of academic and career support for the school.

2.5.2 Initiatives Taken

As part of the effort to support students at Navodaya School, Kadadanamari, several key initiatives were taken to enhance academic performance and create awareness about higher, technical, and vocational education opportunities. The first step was to organize academic support sessions focused on core subjects such as Mathematics, Science, and English. These sessions were designed to help students strengthen their conceptual understanding, clarify common doubts, and practice solving problems based on previous year question papers. The goal was to boost confidence among students preparing for board examinations and improve their overall performance.

To complement the academic support, a series of career guidance sessions were held to educate students about the wide range of educational pathways available after completing school. These included options such as intermediate education, polytechnic diplomas, ITI (Industrial Training Institutes), and professional degree programs in engineering, medicine, and commerce. Students were also introduced to national skill development initiatives and vocational training courses that offer practical career options. Special focus was placed on entrance exams, eligibility criteria, and how to begin early preparation

Another important initiative involved raising awareness about government-funded scholarship programs and financial aid portals like the National Scholarship Portal (NSP) and Vidya Lakshmi. The students were guided on how to find and apply for these opportunities, which can help reduce

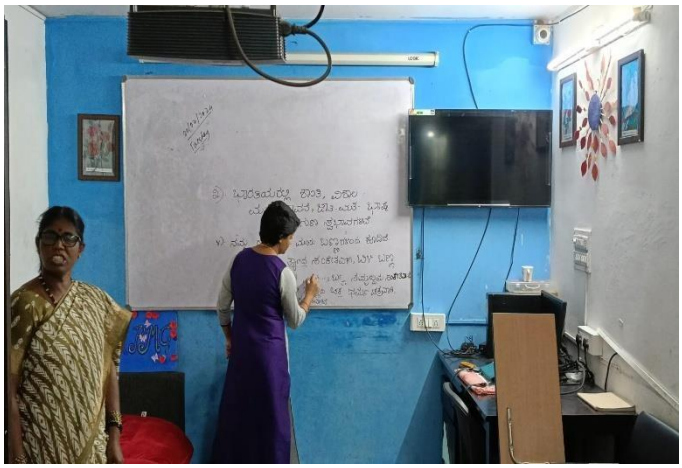
the financial burden of continuing education. This was especially beneficial for students from economically disadvantaged backgrounds who may otherwise lack access to higher studies.

In addition, interactive sessions were conducted on effective exam preparation strategies, including time management, stress control, and daily study planning. Students were taught how to make use of model papers, focus on high-weightage topics, and approach exams with a calm and structured mindset. Motivational talks featuring real-life examples of individuals who succeeded in various professional fields despite challenges were also part of the initiative, aiming to inspire students to believe in their potential.

To ensure personalized support, open Q&A rounds and one-on-one mentoring were arranged for students who required additional guidance or had specific doubts. Furthermore, teachers were engaged and encouraged to continue these efforts after the program through follow-up activities and the use of shared study materials. These combined initiatives created a supportive and encouraging academic environment, promoting both improved results and greater ambition among students to pursue higher education and skilled careers.

Chapter 3

Documentation Of Certificates



CHAPTER 4
OVERALL LEARNINGS

4.1 Innovative approaches taken

Engaging in a variety of socially and technically relevant activities allowed me to step out of my academic routine and immerse myself in real-world problems and community needs. These experiences greatly enhanced my awareness, built confidence, and sharpened both my technical and interpersonal skills. They helped me develop a practical understanding of how theoretical knowledge can be translated into community service and national development efforts.

4.2. Expanded Knowledge and Understanding

Through participation in initiatives like sustainable water management, digital literacy promotion, and school support programs, I developed a strong understanding of India's key developmental challenges — including education gaps, poor digital access, environmental mismanagement, and sanitation issues. I realized how even small actions, such as educating someone on using UPI apps or guiding a student toward a scholarship, can have long-term ripple effects on families and communities.

I also gained valuable knowledge about national missions such as Digital India, Swachh Bharat Abhiyan, Skill India, Jal Shakti Abhiyan, PMKVY, and the New Education Policy (NEP). Understanding how these initiatives are implemented at the ground level gave me insights into public policy, governance, and the responsibilities of citizens, especially youth, in supporting nation-building.

4.3. Research and Preparation

Before each activity, I engaged in topic-specific research to ensure accuracy and relevance in what I presented. For instance, while conducting sessions on cashless transactions, I had to understand how platforms like BHIM, UPI, and mobile wallets worked, what security protocols existed, and how to explain these features to people unfamiliar with digital tools. Similarly, while discussing higher

education opportunities, I compiled detailed information on career options, government scholarships, and vocational courses to guide students effectively.

In projects like garbage disposal system design or water conservation, I explored sustainable practices such as segregation at source, composting, rainwater harvesting, and drip irrigation. This helped me understand environmental engineering at a practical level and how basic technologies can be applied in rural or urban settings.

4.4 Challenges Encountered

A wide range of challenges arose throughout the course of these projects. These included language barriers, lack of student engagement, technological limitations, resource shortages, and even resistance to change from the community. In many cases, explaining concepts like UPI or online portals was difficult for those unfamiliar with smartphones. Likewise, keeping students interested in educational sessions demanded innovative and interactive methods.

Another major hurdle was limited infrastructure — such as schools without proper digital access, or water management programs that lacked proper tanks, pipes, or support from authorities. These challenges tested my patience, flexibility, and ability to think critically under pressure.

4.5 Solutions and Innovations

To overcome these issues, I adapted my approach by simplifying technical content and using local language where needed. I relied heavily on visual aids, real-life stories, hands-on demonstrations, and group activities to make learning interactive and relatable. In classrooms, I used simple games, problem-solving challenges, and storytelling to keep students engaged.

In projects involving awareness (like waste segregation or water usage), I used posters, hand-drawn charts, and live examples to convey ideas effectively. I also collaborated with teachers, village elders, and peers to gain trust and improve communication with the community.

These efforts taught me that creativity, empathy, and collaboration are key tools in overcoming real-world obstacles.

4.6 Personal and Professional Growth

These experiences shaped my character and improved my soft skills like communication, leadership, empathy, and teamwork. I became more confident in public speaking, better at organizing and leading sessions, and more aware of how to work with diverse groups of people. I also realized the importance of research-based preparation, continuous learning, and adaptability.

From a technical perspective, I gained exposure to sustainable practices, digital tools, public outreach techniques, and basic management strategies — all of which are useful in my future career as an engineer or problem-solver.

Most importantly, these activities instilled a deep sense of social responsibility, making me more committed to contributing positively to my community and society at large. I now understand that real learning happens when knowledge is used to solve problems, uplift others, and build a more inclusive and sustainable future.

CHAPTER 5
CONCLUSION

Participating in a wide range of social, environmental, educational, and digital outreach initiatives has been a highly enriching and eye-opening experience. These activities gave me an opportunity to step beyond the classroom and actively contribute to society, applying what I've learned in practical and meaningful ways. They helped me understand how deeply interconnected our communities are, and how even a single effort—when done with sincerity—can inspire change and uplift lives.

One of the most valuable outcomes of this journey was the realization that true education is holistic. It is not confined to academic achievements alone but extends to values like empathy, responsibility, adaptability, and active citizenship. Whether it was guiding a student toward a better academic path, demonstrating a digital payment method to someone unfamiliar with technology, or helping implement eco-friendly practices, every small action carried the potential to make a lasting impact.

These experiences also sharpened my critical thinking and leadership abilities. I learned how to assess situations quickly, identify local challenges, and adapt my strategies based on the audience, environment, and available resources. Facing difficulties—such as lack of infrastructure, low awareness, or resistance to change—taught me patience, perseverance, and the importance of creative problem-solving. It also instilled in me the belief that collaboration, compassion, and communication are key to driving sustainable development.

From a personal growth perspective, these activities strengthened my self-confidence, public speaking, planning, and teamwork skills. I also gained practical knowledge in areas like sustainability, digital tools, and public service, which are directly relevant to my career goals. This journey reaffirmed my desire to use my education and skills to serve the community and contribute to national progress.

In conclusion, these activities have transformed my outlook, broadened my horizons, and shaped me into a more responsible individual ready to take on real-world challenges with commitment and confidence.